

ENGR 3323: Signals and Systems

HW 6_Ch4 Answer Keys

1)

$$\text{a) } y(t) = \underbrace{(2 + 5t)e^{-2t}}_{y_{\text{zir}}(t)} + \underbrace{te^{-2t}}_{y_{\text{zsr}}(t)}$$

$$\text{b) } y(t) = \underbrace{[\sqrt{2}e^{-3t} \cos(4t - \frac{\pi}{4})]}_{y_{\text{zir}}(t)} + \underbrace{[2 + 5.154e^{-3t} \cos(4t - 112.83^\circ)]}_{y_{\text{zsr}}(t)}$$

2)

$$\text{a) } H(s) = \frac{2s - 1}{s^2 + 3s + 2}$$

$$\text{b) } h(t) = [-3e^{-t} + 5e^{-2t}]u(t)$$

$$\text{c) } y_{\text{zir}}(t) = -4e^{-t}u(t) + e^{-2t}u(t).$$

$$\text{d) } y_{\text{zsr}}(t) = \left[-\frac{1}{2} + 3e^{-t} - \frac{5}{2}e^{-2t}\right]u(t)$$

3) At $t=0$, the inductor current $y_1(0) = 4$ and the capacitor voltage is 16 volts. After $t = 0$, the loop equations are

$$\text{a) } 2\frac{dy_1}{dt} - 2\frac{dy_2}{dt} + 5y_1(t) - 4y_2(t) = 40$$

$$-2\frac{dy_1}{dt} - 4y_1(t) + 2\frac{dy_2}{dt} + 4y_2(t) + \int_{-\infty}^t y_2(\tau) d\tau = 0.$$

$$\text{b) } y_1(t) = [8 + 17.89e^{-1.5t} \cos(\frac{t}{2} - 26.56^\circ)]u(t)$$

$$y_2(t) = 20\sqrt{2}e^{-1.5t} \cos(\frac{t}{2} - \frac{\pi}{4})u(t).$$

4)

$$\text{a) i) } y_1(t) = [6 + 9.22e^{-t} \cos(2t - 130.6^\circ)]u(t)$$

$$\text{ii) } y_2(t) = \frac{1}{10}\{6 + 9.22e^{-(t-5)} \cos[2(t-5) - 130.6^\circ]\}u(t-5)$$

$$\text{b) } \ddot{y}(t) + 2\dot{y}(t) + 5y(t) = 2\dot{x}(t) + 3x(t)$$

5)

$$\text{a) } y_{\text{ss}}(t) = 7.5u(t)$$

$$\text{b) } y_{\text{ss}}(t) = \frac{\sqrt{13}}{8} \cos(2t + 3.69^\circ)u(t)$$

$$\text{c) } y_{\text{ss}}(t) = \frac{\sqrt{18}}{13} \sin(3t - 112.62^\circ)u(t)$$

$$\text{d) } y_{\text{ss}}(t) = \frac{\sqrt{18}}{13} e^{j[3t-67.62^\circ]}u(t)$$